$$\int_0^\infty x^2 e^{-ax}\,dx = \frac{2}{a^3} \qquad \int \frac{dx}{\sinh^2 x\cosh^2 x} = -2\coth 2x \qquad \int_0^\infty \frac{\cos ax}{\sqrt{a^2+x^2}}\,dx = K_0(ua) \qquad \int_0^\infty x e^{-(x^2)}\,dx = \frac{1}{2r^2} \qquad \int_0^\infty \frac{1-\cos ax}{x^2}\,dx = \frac{\pi|a|}{2}$$

$$\int_0^{\frac{\pi}{2}} \frac{\sinh x}{\sin x}\,dx = \frac{\pi}{2}$$

$$\int \tanh x\,dx = \ln\cosh x \qquad \int_0^\infty \frac{x\sin ax}{a^2+x^2}\,dx = \frac{\pi}{2}e^{-ax} \qquad \int \frac{\sinh^{p-1} x}{\cosh^p x}\,dx = \frac{\cosh^{p-1} x}{p-1} \qquad \int_0^\infty \frac{dx}{e^{ax}-1} = \infty \qquad \int_0^\infty \frac{x^{p-1}}{1+x^q}\,dx = \frac{\pi}{q\sin\frac{p\pi}{q}} \qquad \int \coth x\,dx = \ln\sinh x$$

$$\int_0^\infty \frac{x^2\,dx}{e^{ax}-1} = \frac{\pi^4}{15a^3} \qquad \int_0^\infty \frac{dx}{e^{ax}+1} = \frac{\ln 2}{a} \qquad \int \coth x\,dx = \ln|\sinh x| \qquad \int_0^\infty \cos(x^2)\,dx = \sqrt{\frac{\pi}{2}} \qquad \int \frac{(\ln x)^2\,dx}{x} = \frac{(\ln x)^3}{3}$$

$$\int \sinh x\,dx = \cosh x \qquad \int \frac{dx}{\sinh^2 x} = -\coth x \qquad \int_0^\infty \frac{x^{m-1}\,dx}{(1+x)^{m+n}} = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} \qquad \int_0^\infty \frac{x^3\,dx}{\sinh ax} = \frac{\pi^4}{30a^4} \qquad \frac{d}{dx}\operatorname{Arsh}\frac{x}{a} = \frac{1}{\sqrt{x^2+a^2}}$$

$$\int x^n \cos x\,dx = x^n\sin x - n\int x^{n-1}\sin x\,dx \qquad \int x\cos x\,dx = \cos x + x\sin x \qquad J_n(x) = \frac{\left(\frac{x}{2}\right)^n}{n!}\left[1 - \frac{\left(\frac{x}{2}\right)^2}{1(n+1)} + \frac{\left(\frac{x}{2}\right)^4}{1\cdot 2(n+1)(n+2)}\right] \qquad \int_0^\infty \frac{\sin px}{\sinh qx} = \frac{\pi}{2q}\operatorname{tg}\frac{\pi p}{2q}$$

$$\frac{d^2u}{dx^2} + \frac{1}{x}\frac{du}{dx} + \left(1 - \frac{n^2}{x^2}\right)u = 0 \qquad \int \operatorname{th} x\,dx = \ln\cosh x \qquad \int \frac{\sinh^2 x}{\cosh x}\,dx = \sinh x - \operatorname{arctg}(\sinh x) \qquad \int \operatorname{th}^2 x\,dx = x - \operatorname{th} x$$

$$\int_0^\infty \frac{dx}{1+2x+x^2} = \frac{1}{2} \qquad \int_0^\infty \frac{x\,dx}{e^{ax}+1} = \frac{\pi^2}{12a^2} \qquad \int_0^\infty \frac{x^3\,dx}{\sinh ax} = \frac{7\pi^4}{30a^4} \qquad \int_0^\infty \frac{x^2\,dx}{\cosh ax} = \frac{\pi^3}{3a^3} \qquad \int \frac{\sinh^2 x}{\cosh^2 x}\,dx = \sinh x + \operatorname{sech} x \qquad \int \frac{x^2\,dx}{(1+x)^2} = \frac{e^x}{1+x}$$

$$\int \frac{dx}{\sinh 2x} = \frac{1}{2}\ln|\tanh x| \qquad \int \frac{dx}{\cosh x+1} = \operatorname{th}\frac{x}{2} \qquad \int \frac{dx}{\sinh^2 x - 1} = \ln\left|\frac{\sinh x - 1}{\sinh x + 1}\right| \qquad \int \frac{dx}{x\ln x} = \ln|\ln x| \qquad \int \frac{(\ln x)^2\,dx}{x} = \frac{(\ln x)^{p+1}}{p+1}$$

$$\int x\cosh x\,dx = x\sinh x - \cosh x \qquad \int \cosh^2 x\,dx = \frac{\sinh 2x}{2} + x \qquad \int \sinh^2 x\,dx = \frac{\sinh 2x}{2} - x \qquad \int \frac{x\,dx}{r} = \frac{x^2}{2} - \frac{a^2}{2}\ln|x+r| \qquad \int \frac{x^2\,dx}{r^2} = r + \frac{a^2}{r}$$

$$X = X_f - X_i \qquad \Delta V = V_f - V_i \qquad a = \frac{\Delta W}{\Delta t}$$

$$V = \frac{\Delta v}{\Delta t} \qquad a = \frac{\Delta v}{\Delta t}, \quad W = \frac{\Delta Q}{\Delta t}, \quad f = \frac{1}{T}$$

$Q_{mt} = \Delta W_{by} + \Delta E$, Q (quant) (const ΔT $G = 6.6\cdot 10^{-11}\ N\,m^2/Kg$

$K + U$, $E_i = E_f$ $\quad \lambda \quad \frac{L}{4}$, $p = mv$

ms^2 $X = A\cos(wt) = \lambda/2$

$= \{or\}\ A\sin(wt)$, $\quad M_t = 5.97\cdot 10^{24}\ Kg$ $\quad Re = E\,37(C)_r$

$v = Aw(\sin wt)\ \{or\}\ (Aw\cos ust)$

$a = Aw^2\cos(wt) = \{or\}\ Aus^2\sin(wt)$

$$\frac{GMe}{Re} = gRe; \quad \frac{GM}{T^2}; \quad \sum m_3\vec{V}_3 = \sum \vec{m_3 V_5}$$

$$\mathcal{L} = \oint E_p t$$

$$\nabla\cdot E = 0 \qquad \nabla\times E = -\frac{1}{c}\frac{\partial H}{\partial t} \qquad \nabla\cdot H = 0 \qquad \nabla\times H = \frac{1}{c}\frac{\partial E}{\partial t}$$

$$f(\omega) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x\omega}\,dx \qquad \frac{dt}{d\theta}$$

$$i\hbar\frac{\partial}{\partial t}\Psi = H\Psi$$

$$\rho\left(\frac{\partial v}{\partial t} + v\,\nabla v\right) = -\nabla p + \nabla\cdot T + f$$

$$\boxed{H = -\sum p(x)\log p(x)}$$

$$\frac{1}{2}\sigma^2 S^2\frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} - rV = 0$$

$$+\sum_{i=1}^{n}\frac{q_i}{x}M^M + c_i\frac{D}{\partial} + c_0 D_i + \frac{Q(p-D)}{2p}M^M + F_0 N + F_0 N + \sum_{i=1}^{n} D_i w_i d_i \frac{(1+\alpha)}{F_r}$$

$$TC(Q, q_i, m_i) = \sum_{i=1}^{n}\left[\frac{D_i}{m_i q_i}S_i + c_i D_i + \frac{q_i M_i^v}{2}\left(m_i\left(1 - \frac{D_i}{P_i}\right) - 1 + 2\frac{D_i}{P_i}\right)\right] +$$

$$\begin{bmatrix} \frac{d\Delta p(s\phi)}{d\phi} \\ \frac{d\Delta M(s,\phi)}{d\phi} \end{bmatrix} = \begin{bmatrix} \gamma & -\mathcal{L} \\ -\beta & 0 \end{bmatrix}\begin{bmatrix} \Delta p(s,\phi) \\ \Delta M(s,\phi) \end{bmatrix}$$

$$\int_0^{\frac{\pi}{2}}(\log\sin x)^2\,dx - \int_0^{\frac{\pi}{2}}(\log\cos x)^2\,dx = \frac{\pi}{2}\left\{\frac{\pi^2}{12} + (\log 2)^2\right\}$$

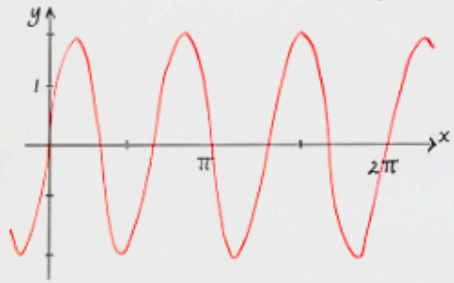
$y = 2\sin 3x$

$A = 2,\ p = \frac{2\pi}{3}$

$y = -2\cos 2t$

$A = 2;\ p = \pi$

$y = 2\cos 3x$



$y = 4 - 4\cos 2(x - \pi)$

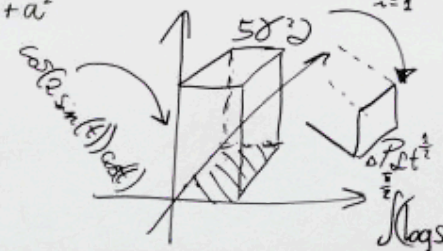
$y + 3 = 6\sin\frac{\pi}{4}(x+2)$

$x^2 + xy + y^2 = 1$

$\cancel{x^2 - 2xy - y^2 = 4}$

$x^2 - 3xy + y^2 = -6$

$$c^2 = (AB)^2 = (b\cos C - a)^2 + (b\sin C - 0)^2$$
$$= b^2\cos^2 C - 2ab\cos C + a^2 + b^2\sin^2 C$$
$$= b^2(\cos^2 C + \sin^2 C) - 2ab\cos C + a^2$$
$$= b^2\cdot 1 - 2ab\cos C + a^2$$
$$= a^2 + b^2 - 2ab\cos C$$

$x^2 - 2xy + y^2 - \sqrt{2}x - \sqrt{2}y = 0$

$\cos x = -\frac{\sqrt{3}}{2}$ $\qquad 5x^2 + 2xy - y^2 = 6$